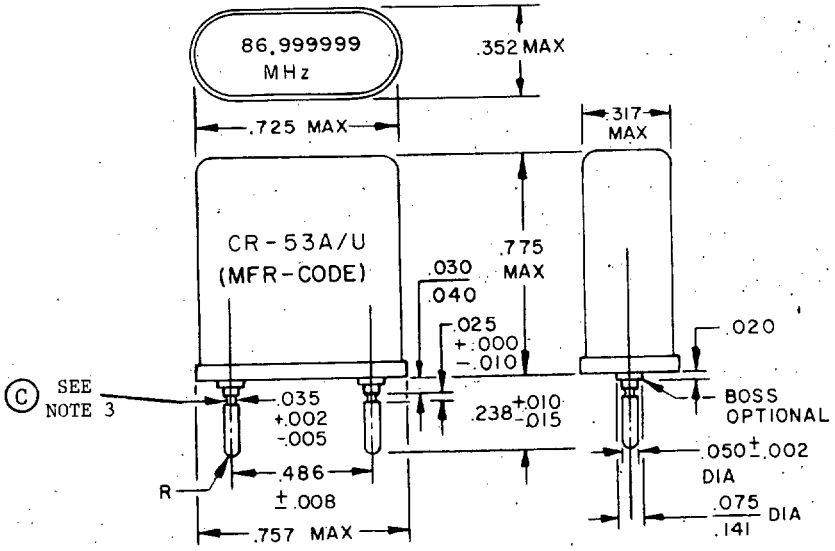
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-53A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of specification MIL-C-3098.

Pertinent characteristics: 50 to 87 MHz; fifth overtone; noncontrolled; series.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.352	8.94
.005	.13	.025	.64	.075	1.91	.486	12.34
.008	.20	.030	.76	.141	3.58	.725	18.42
.010	.25	.035	.89	.238	6.05	.757	19.23
.015	.38	.040	1.02	.317	8.05	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
Ⓒ 3. The pin undercut may be omitted.

FIGURE 1. Dimensions and configuration.

Ⓒ denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (MIL-H-10056/2)
Dimensions, marking, and configuration	See figure
Frequency range	50 to 87 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Mode of oscillation	Fifth mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	20.0 ±4.0 mW
Calibration values:	
Equivalent resistance	60 ohms
Resistor voltage drop	1.1 volts
Operating temperature range	-55° to +105° C, inclusive
© Frequency tolerance	±50 PPM
Equivalent resistance	60 ohms, maximum
Shock:	
© Frequency change permitted	±20 PPM
Equivalent resistance	Shall not exceed 60 ohms
Vibration: 1/	
© Frequency change permitted	±20 PPM
Equivalent resistance	Shall not exceed 60 ohms
© Thermal shock:	
Frequency change permitted	±20 PPM
Equivalent resistance	Shall not exceed 60 ohms
Aging:	
© Frequency change permitted	±5 PPM

1/ Method 201 of MIL-STD-202.

NOTES:

© 1. The socket designed to mate with crystal holder HC-6/U is military part no. M12883/04-01 (see MIL-S-12883/4). For Navy use, other sockets listed in MIL-STD-242 may be used.

2. This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army -
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5955-0512-9)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3098/31C CRYSTAL UNIT, QUARTZ, CR-53A/U

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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A. GIVE PARAGRAPH NUMBER AND WORDING

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